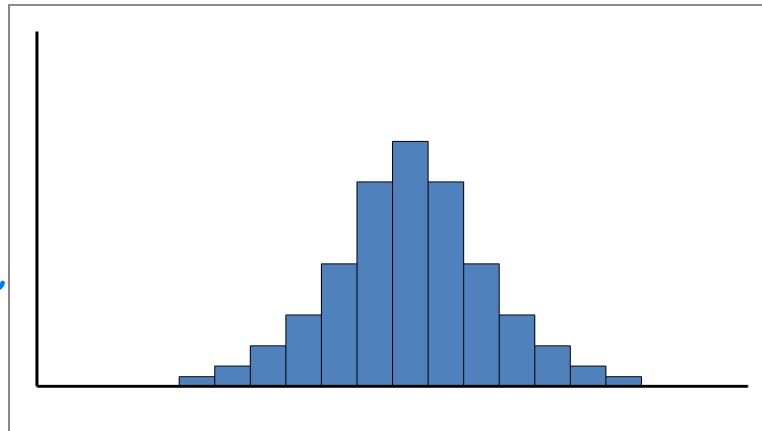
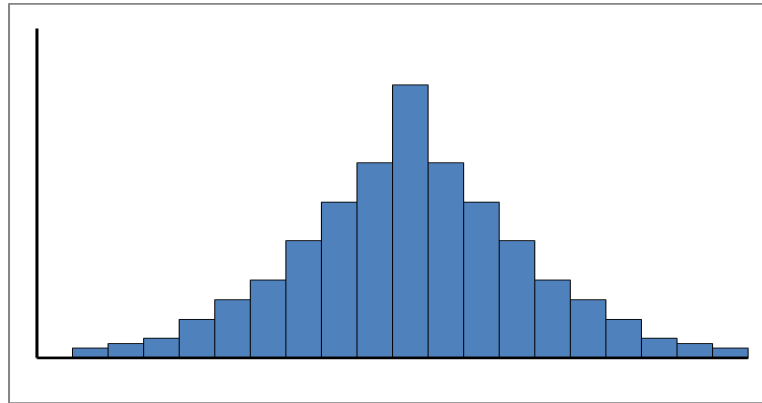


SUMMARY MEASURES: DISPERSION MEASURES

Motivation

Consider the two distributions below.



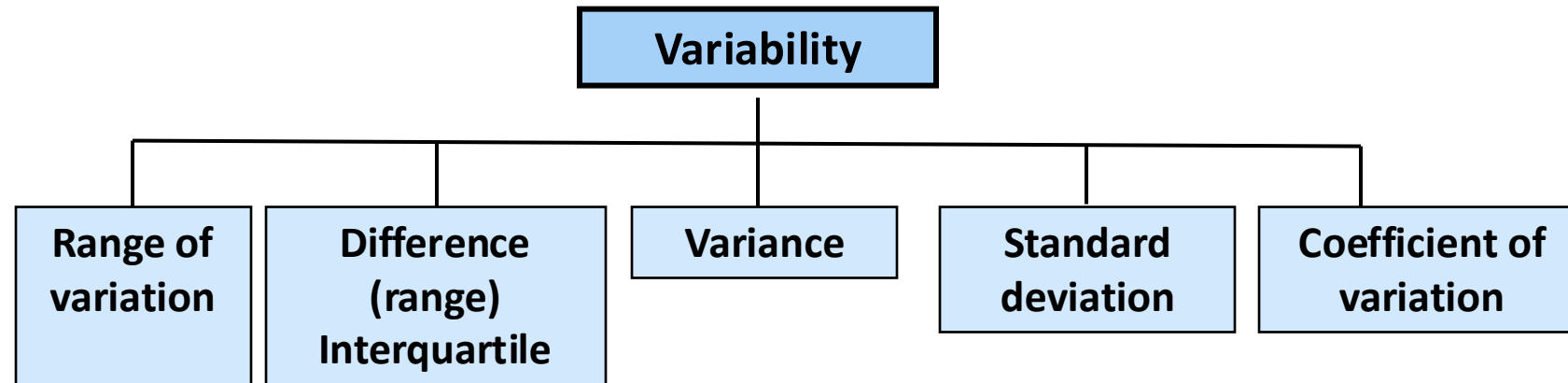
Considerations?

- ▶ Both distributions are symmetrical and are characterised by the same measures of central tendency
- ▶ The level of dispersion around the centre is different: the first distribution is more dispersed (longer tails)
- ▶ We can effectively describe and summarise the differences between the distributions using a box plot or by comparing their percentiles
- ▶ However, we may be interested to use **measures** that effectively summarise the differences between the two distributions in terms of dispersion

Introduction

A **measure of dispersion or variability** quantifies and summarises the amount of dispersion in the observed data

- ▶ Quantifying the amount of dispersion is crucial especially when summarising the distributions of numerical variables.
- ▶ Although measures of dispersion have also been introduced for qualitative variables, here we focus on the most relevant and common measures of variability, defined only for numeric variables



Variability/dispersion measures

Range of variation and interquartile range

Range of variation (or range)

The simplest and most intuitive measure of dispersion is based on the evaluation of the **width of the interval** containing **all the observed data**, the so-called **range of variation**, given by the **difference between the highest and lowest observed value**

- ▶ **Example.** Amount spent on the website of a pharmacy (11 transactions in a certain day)

Ordered data	Range
55.5 60.4 61.3 62.3 66.7 72.8 74.6 80.2 81.1 90.5 91.3	91.3 - 55.5 = 35.8
	$\text{MAX} - \text{MIN} = \text{RANGE}$

- RANGE is not robust wrt extreme values
 ↓ changes in extreme values can impact the range a lot

Range of variation (or range)

Example. Amount spent on a pharmacy's website related to 9 transactions in one day

Ordered data

55.5 60.4 62.3 66.7 72.8 74.6 81.1 90.5 91.3

Range

91.3 - 55.5 = 35.8

Suppose that exactly the same data were observed on other days, except for the maximum or minimum transaction amount

Ordered data

55.5 60.4 62.3 66.7 72.8 74.6 81.1 90.5 **111.1**

55.5 60.4 62.3 66.7 72.8 74.6 81.1 90.5 **148**

25.8 60.4 62.3 66.7 72.8 74.6 81.1 90.5 91.3

60.3 60.4 62.3 66.7 72.8 74.6 81.1 90.5 91.3

Range

55.6

92.5

65.5

30.0

*Being based exclusively on minimum and maximum values, the **range** is a **non-robust** measure, very sensitive to the presence of **extreme values**, very far from the rest of the data.*

The interquartile range (or difference)

The range is sensitive to extreme values i.e. **not robust** with respect to extreme values.

A possibility to circumvent this problem is to eliminate the largest and the smallest values and to calculate the range on the remaining ones.

The **interquartile range** is the length of the interval containing the 50% of central values, i.e. the range calculated by excluding 25% of the smallest data and 25% of the largest data. It is defined as the **difference between the third and first quartile**: $IQR = Q_3 - Q_1$ (width of the box in the box plot), and inform about the width of the interval including central values around the median. It is a **robust measure, not influenced by extreme values**.

- ▶ **Example.** Amount spent on the website of a pharmacy (9 transactions in one day)

Ordered data

55.5 60.4 62.3 66.7 72.8 74.6 81.1 90.5 91.3

↑ ↑

$Q_1 = 62.3$ $Q_3 = 81.1$

IQR

$81.1 - 62.3 = 18.8$

$Q_3 - Q_1 = IQR$

Variability/dispersion measures

Variance and standard deviation

The variance: deviations from the mean

The **variance** is a measure of the dispersion of data around their mean.

Referring initially to a sample, consider:

x_i value for the i -th observation

$\bar{x}$ sample mean $\longrightarrow \bar{x}$: centre of the distribution

The **deviation** of x_i from the mean:

$(x_i - \bar{x})$: deviation of x_i from $\bar{x}$

- ▶ quantifies the deviation of the i -th observation from the centre (i.e. the sample mean)
- ▶ quantifies the error that occurs if x_i was replaced (or estimated) by the sample mean

A measure of dispersion can therefore be based on the deviations from the mean. Such a measure would provide both a summary of the dispersion around the centre and a measure of the risk involved when using the mean as “representative” of the observed data.

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n \bar{x} = 0$$

collider



The variance: summary of deviations

Note that

- **The sum of the deviations of the data from the mean is zero** (the mean is the centre of gravity of the data). Thus, the deviations from the mean cannot be directly used to assess dispersion.
- When assessing the dispersion around the mean, we are not really interested to the **direction of the deviations** (values higher or lower than the mean) but, rather, to their **magnitude**

The dispersion around the mean can therefore be summarised by the mean of the squared deviations from the mean, the so-called **variance**

The variance: definition for population and sample

The definition of variance is slightly different for populations and samples.

- ▶ **The population variance** (variance of a population) is defined as:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \quad \text{where } \mu \text{ is the population mean}$$

- ▶ **The sample variance** is instead defined as:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{where } \bar{x} \text{ is the sample mean}$$

This is not exactly the **mean of the squared deviations**, since the sum is divided by $(n - 1)$ and not by n . The reason for this difference will be explained when discussing about inference, and specifically about estimation

VARIANCE OF
A RANDOM VARIABLE

$$\text{Var}(X) = E[(X - E[X])^2]$$

trust me
for now ♥

(wait lecture 13/14)

The variance: characteristics

Without loss of generality we focus on the **sample variance**:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

because the square of the deviation is considered

unlike RANGE and IQR

note that

- ▶ The variance is calculated based on **all the observed values** and not focusing on specific values or summaries as in the case of the range and interquartile range
- ▶ Being based on the mean (that is sensitive to extreme values) and being itself the mean of the squared deviations of the data from mean, the variance is a **non-robust measure** of dispersion. Thus, it is influenced by possible extreme values (even more than the sample mean)
- ▶ The variance can be interpreted as the “average” **squared error** incurred when replacing the raw data with their mean. Therefore, it can also be regarded as a measure of the **reliability of the mean as a summary of the data**

PROOF OF INTEREST CALCULATION FORMULA (optional proof, but good practice)

SAMPLE VARIANCE

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} \cdot \frac{1}{n} \sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$

$$= \frac{n}{n-1} \left(\frac{1}{n} \sum x_i^2 - 2 \sum \frac{x_i \bar{x}}{n} + \frac{1}{n} \sum \bar{x}^2 \right)$$

correction factor

$-2\bar{x} \frac{\sum x_i}{n} = -2\bar{x}^2$

$\rightarrow = \bar{x}^2$

$$= \frac{n}{n-1} \left(\frac{1}{n} \sum x_i^2 - \bar{x}^2 \right)$$

mean of the squares

square of the mean

The variance: indirect calculation formula [OPTIONAL]

Focusing on the **sample** variance, we note that:

$$\begin{aligned}s^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i^2 + \bar{x}^2 - 2x_i\bar{x}) = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2\bar{x} \sum_{i=1}^n x_i \right] \\&= \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2n\bar{x}^2 \right] = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right] \\&= \frac{n}{n-1} \left[\sum_{i=1}^n \frac{x_i^2}{n} - \bar{x}^2 \right]\end{aligned}$$

The variance can be calculated from the mean of the squared data and the square of the mean!

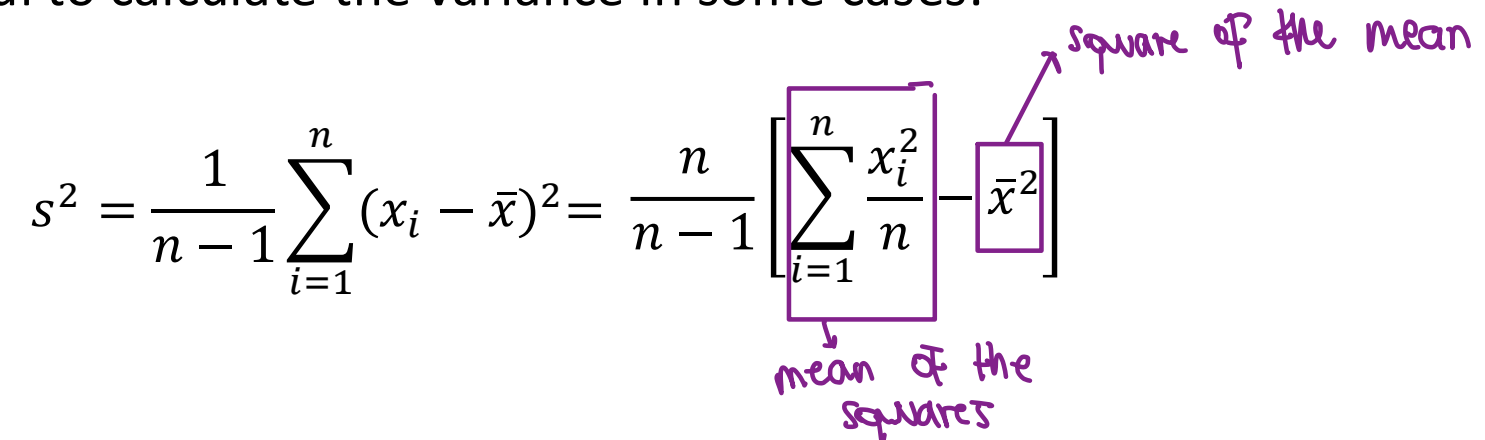
In the case of a **population**, the same reasoning leads to:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \sum_{i=1}^N \frac{x_i^2}{N} - \mu^2$$

The variance: indirect calculation formula

Focusing on the **sample** variance, we introduce the so-called indirect calculation formula, that can be useful to calculate the variance in some cases:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} \left[\sum_{i=1}^n \frac{x_i^2}{n} - \bar{x}^2 \right]$$



The variance can be calculated based on the mean of the squared data and on the squared mean!

CASES ① raw data ② freq. table ③ freq. table with interval classes

We will use this formula only in particular cases, and specifically when the variance is calculated based on grouped data (for example in the case of variables measured in classes).

METHODS ① indirect calculation formula ② distr. summary. x (raw)

The variance: calculation on raw data [Case 1: raw data]

Example. Age in years measured on a sample of $n = 9$ customers: **35 25 54 60 44 61 27 55 62**

- ▶ *The average is:*

$$\bar{x} = \frac{35 + 25 + \dots + 55 + 62}{9} = \frac{423}{9} = \mathbf{47}$$

- ▶ *The variance is:*

$$s^2 = \frac{(35 - 47)^2 + (25 - 47)^2 + \dots + (55 - 47)^2 + (62 - 47)^2}{8} = \frac{1740}{8} = \mathbf{217.5}$$

$\swarrow$
 $n-1$

- ▶ *Indirect calculation formula:*

$$\frac{\sum_{i=1}^9 x_i^2}{9} = \frac{35^2 + 25^2 + \dots + 62^2}{9} = \frac{\mathbf{21621}}{9} = \mathbf{2402.333} \Rightarrow s^2 = \frac{9}{8} (2402.333 - 47^2) = \mathbf{217.5}$$

$$\frac{n}{n-1} \left(\frac{1}{n} \sum x_i^2 - \bar{x}^2 \right)$$

Standard Deviation

The interpretation and the evaluation of the variance can be complicated by the fact that, being based on the **squared deviations from the mean**, its unit of measurement is the squared unit of measurement of data.

→ DON'T CARE

- ▶ A measure of dispersion around the mean having the same unit of measurement as the data is obtained by considering the square root of the variance, the **standard deviation (sd)**:

Sample Standard Deviation:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Population Standard Deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

- ▶ The standard deviation can be interpreted as a measure of the **average (standard) distance (absolute deviation) of the data from the mean** (since the variance is the average squared deviation from the mean)

GOOD! (IMPORTANT)

Rstudio: `distr.summary.x()` - dispersion

The function `distr.summary.x()` allows determining dispersion measures, specified via the `stats` argument. You can either use `stats="dispersion"` to obtain all dispersion measures, or list the specific statistics you are interested in (`"range"`, `"IQRrange"`, `"sd"`, `"var"`)

```
distr.summary.x(x, stats="dispersion", data)
distr.summary.x(x, stats=c("range", "IQRrange", "sd", "var"), data)
```

```
> distr.summary.x(x=Profitability, stats="dispersion", data=Loyalty)
Summary measures for Profitability
Measures of dispersion
```

n	n.a	range	IQRrange	sd	var	cv
125	0	765.2	141.3	156	24335.64	0.67

We will discuss about this in a moment

The dispersion measures suggest that the distribution has relatively long tails, as one can note from the difference between the range and the interquartile range. The standard deviation indicates that the deviation (in absolute value) of the data from the mean is 'on average' 156 points. As we will discuss in more details later, it is not trivial to assess whether the observed standard deviation is high or not, since such evaluation must necessarily take into account also the size of the mean around which the data are dispersed

Variance and standard deviation for grouped data

→ definition $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \sum_{k=1}^6 (x_k^* - \bar{x})^2 \cdot f_k$

Nr_visits (**Loyalty** dataframe): number of visits to the store in the considered period

Nr_visits	f_k	p_k	$x_k^* p_k$	$x_k^{*2} p_k$
$x_1^* = 1$	18	0.144	0.144	0.144
$x_2^* = 2$	48	0.384	0.768	1.536
$x_3^* = 3$	35	0.280	0.840	2.520
$x_4^* = 4$	16	0.128	0.512	2.048
$x_5^* = 6$	6	0.048	0.288	1.728
$x_6^* = 7$	2	0.016	0.112	0.784
Totale	125	1	2.664	8.760

Variance and standard deviation from the frequency table?

To calculate variance and standard deviation, one must calculate the mean and the 'average' of the squared deviations from it or, using the indirect calculation formula, the mean of the squared values

In the case of grouped data, we saw that the **mean** can be calculated as **the sum of the observed distinct values weighted by their relative frequency**.

In this case, the mean is 2.664

Using the same procedure, the **mean of the squared observed values** can be calculated obtaining **8.760**

This gives $s^2 = (125/124)[8.760 - 2.664^2] = 1.6765$ and $s = 1.295$

$\frac{1}{n/n-1} (\sum (x_k^*)^2 p_k - \bar{x}^2)$ indirect calculation formula with a freq. table ← REMEMBER THIS

Variance and standard deviation for grouped data

In the case of discrete variables, the variance can be calculated also for grouped data, that is when only the variable's frequency distribution is available, based on the observed distinct values and on their relative (or absolute) frequencies.

For a **sample**

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \sum_{k=1}^K (x_k^* - \bar{x})^2 f_k = \frac{n}{n-1} \sum_{k=1}^K (x_k^* - \bar{x})^2 p_k$$

$$= \frac{n}{n-1} \left[\sum_{k=1}^K x_k^{*2} p_k - \bar{x}^2 \right] \leftarrow \text{remember}$$

Calculation for Variables measured in interval classes

Case 3:
freq. table

with interval
classes

Example: variable **Income** (dataframe **Loyalty**), collected in interval classes

Income	f_k	p_k	m_k	$m_k p_k$	$m_k^2 p_k$
[5,15)	14	0.112	10	1.12	11.20
[15,20)	26	0.208	17.5	3.64	63.70
[20,25)	37	0.296	22.5	6.66	149.85
[25,35)	24	0.192	30	5.76	172.80
[35,60)	17	0.136	47.5	6.46	306.85
[60,80]	7	0.056	70	3.92	274.40
Totale	125	1.00		27.56	978.80

Variance and standard deviation?

In this case, it is not possible to calculate the measures exactly; they can only be approximated **by discretizing the variable on the midpoints of the intervals** and proceeding as just seen for the grouped data

Discretizing on the midpoints of the intervals

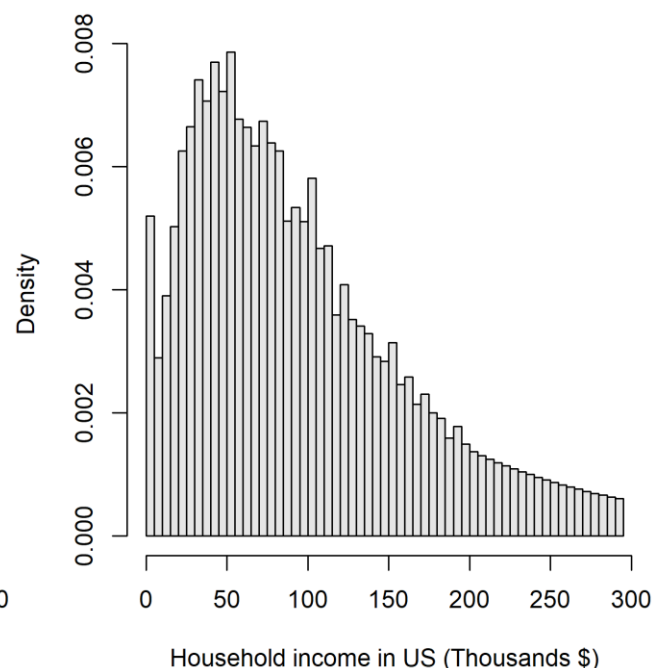
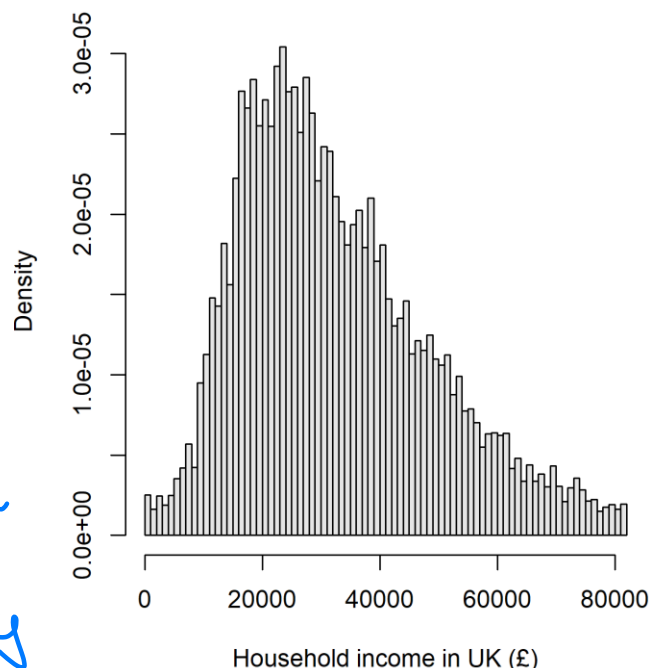
- ▶ The average is **27.56**
- ▶ The mean of the squared values is **978.80**

This gives $s^2 = (125/124)[978.80 - 27.56^2] = 221.01$ e $s = 14.87$ ← approximation

$$\left(\frac{n}{n-1} \right) \left(\sum m_k^2 p_k - \bar{x}^2 \right) \leftarrow \text{remember this}$$

Applications / 1

Comparison of income per household in UK and US (excluding the richest 5%, US data processed for teaching purposes)



Variance and standard deviation **both have a unit of measurement**; therefore they are **absolute** measures of dispersion.
It is meaningless to compare the extent of dispersion in the two distributions based on these measures!

UK: $\bar{x} = 32467.03$; $s = 16190.32$

US: $\bar{x} = 94.88$; $s = 66.29$

Can we conclude that the distribution of income in the UK is more dispersed than in the US?

to compare relative variability of two distributions we divide

s by $|\bar{x}|$: we obtain CV

std: average distance of obs from the mean
- generally most obs are within 2 or 3 std from the mean
- generally obs more than 4 or 5 std from the mean are rare

Variability/dispersion measures

The coefficient of variation

The coefficient of variation

The coefficient of variation is defined as:

$$CV = \frac{s}{|\bar{x}|} \quad \bar{x} \neq 0$$

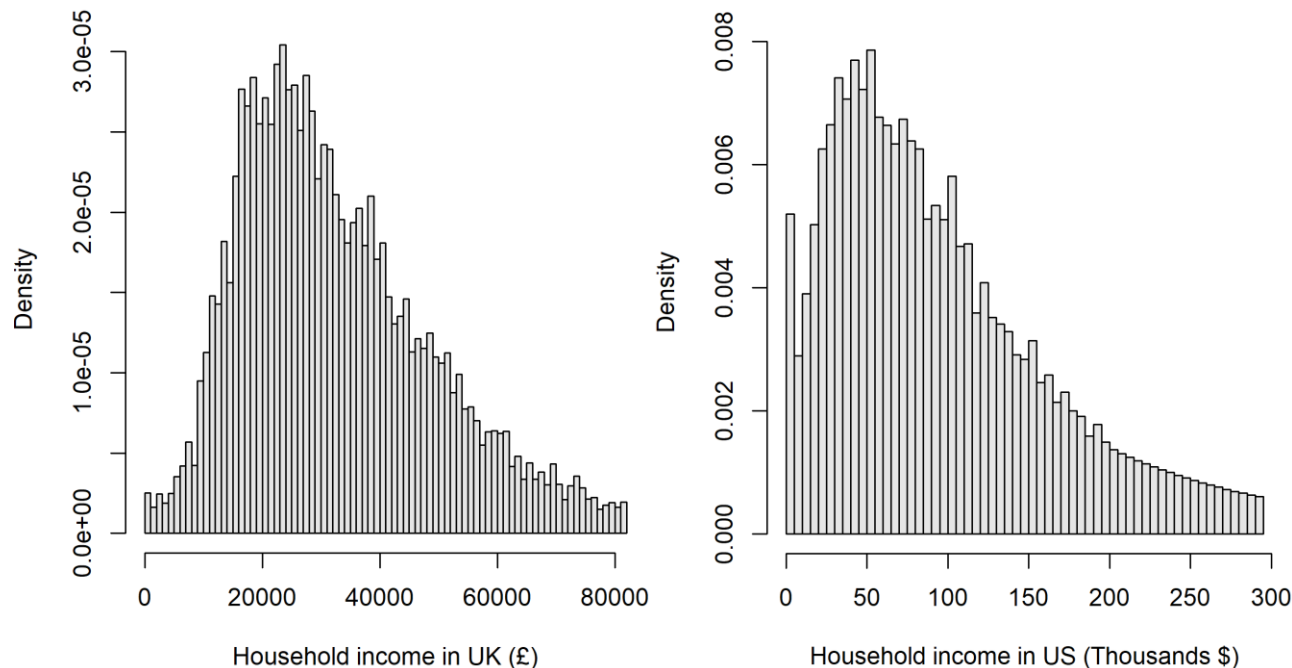
careful! Don't use the CV even when $\bar{x}$ is close to zero as it becomes a very unstable measure

It measures the amount of dispersion in the data in relation to their mean.

- ▶ It is an a-dimensional index: it has no unit of measurement. Consequently, it can (and should) be used to ¹compare the dispersion of variables with different units of ²measurement or with the same unit of measurement but means structurally very different (for example, the price of cars and the price of scooters)
- ▶ However, the coefficient of variation does **not have a well-defined range**; therefore, it is not possible to draw conclusions on the level of dispersion of a single distribution based on the CV

Applications / 1

Comparison of income per household in UK and US (excluding the richest 5%, US data processed for teaching purposes)



UK: $\bar{x} = 32467.03$; $s = 16190.32$ (sterling, £)

US: $\bar{x} = 94.88$; $s = 66.29$ (thousands, \$)

Can we conclude that the distribution of income in the UK is more dispersed than in the US?

The CV for UK and US is

CV_UK: $16190.32/32467.03 = 0.499$

CV_US: $66.29/94.88 = 0.699$

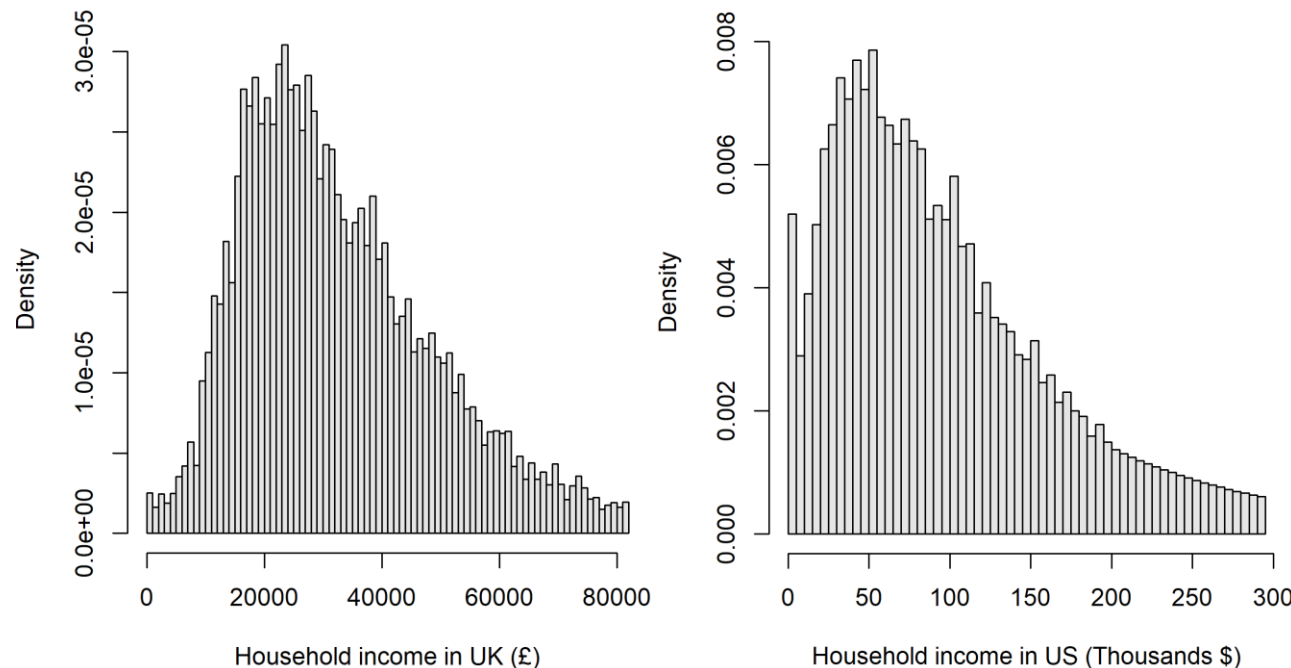
In relation to the mean, the income is more dispersed in the US, despite lower standard deviation and variance:

- 1) due the different **currency** (£ and \$)*
- 2) because the US income is **measured** in thousands of \$*

The CV allows a correct comparison between data with different units of measurement.

Applications / 1

Comparison of income per household in UK and US (excluding the richest 5%, US data processed for teaching purposes)



The CV of the distribution of income in US is [data in thousands of dollars]:

$$CV = 66.29 / 94.88 = 0.699$$

[data in dollars]:

$$CV = 66290 / 94880 = 0.699$$

*By changing the scale (dollars or thousands of dollars), the mean, the variance and the standard deviation **change!***

The coefficient of variation instead does not change because it does not react to changes in the unit of measurement!!!

US (income in **thousands of dollars**): $\bar{x} = 94.88$; $s = 66.29$

US (income in dollars): $\bar{x} = 94880$; $s = 66290$

The mean and the standard deviation change with the unit of measurement

Applications / 2

Let us consider the average one-year return and volatility (standard deviation) of the shares of some companies, evaluated considering the same years

	Mean	Variance	sd
Amazon	0.362	0.471	0.686
Coca-Cola	0.072	0.023	0.151
Colgate	0.075	0.019	0.137

	CV
Amazon	1.896
Coca-Cola	2.109
Colgate	1.824

Comments?

Amazon's shares have a much higher volatility compared to those of Coca-Cola and Colgate.

The standard deviations of the returns of the three stocks are different, but so are their averages! The average returns of Coca-Cola and Colgate are lower, and this implies lower deviations from their averages.

*In contrast, the CVs show that **the variability of returns, expressed as a % of the average return, is greater for Coca-Cola! The conclusion is opposite to that reached by comparing the standard deviations.***

The CV also allows a correct comparison between data with the same unit of measurement (here %) but very different averages (related to different phenomena)